

Arc Reactor Mining Lab

Safe multi-coin mining plan for operator-owned hardware, receive-only wallets, and auditable expense tracking.

No private keys. No brute force. No hidden mining. No Lantern cash-ledger reuse as crypto balance.

This package documents how to test CPU, GPU, and ASIC mining lanes without giving any tool custody over funds.

Deliverable	File
Operator guide	docs/arc-reactor-mining-lab/README.md
Workbook	data/arc-reactor-mining-lab/arc_reactor_mining_lab_workbook.xlsx
Templates	data/arc-reactor-mining-lab/*.csv and receipt-ledger-template.jsonl
Sample configs	configs/arc-reactor-mining-lab/
Validation	scripts/validate_arc_reactor_mining_lab.py

Executive Decision

- Use Monero as the CPU learning lane, because RandomX is CPU-optimized and discourages specialized hardware.
- Use Ravencoin as the GPU learning lane only if the operator already owns GPUs and can manage heat.
- Treat Kaspa as ASIC-only for this lab. Do not buy an ASIC from static estimates; model already-owned hardware first.
- Use current calculators and actual wall power for every run. This document deliberately avoids fixed profit promises.

Safety Model

The lab separates wallet accounting from mining execution. Miners are configured with receive addresses only. The local Lantern-style ledger records receipts and off-ramp events after they happen; it is not a crypto wallet, custodian, balance oracle, or private-key store.

Boundary	Allowed	Not allowed
Wallets	Receive addresses and public transaction IDs	Seed phrases, private keys, exchange passwords
Hardware	Owned CPU/GPU/ASIC devices with measured wall power	Unapproved computers, cloud abuse, covert usage
Software	Official/open-source miners and documented pool mining	Malware, key extraction, brute forcing
Accounting	Append-only receipt rows and exchange receipts	Invented balances or unverified cash claims

Hardware Intake

Inventory begins with CPU/GPU/ASIC identity, hashrate, power draw, wall measurement, operating hours, electricity rate, and cooling readiness. The workbook includes formulas for power cost so the operator can adjust assumptions without editing code.

Device	Type	Lane	Watts	Hours	kWh rate	Notes
cpu-01	CPU	RandomX	65	4	0.182	Replace example row with measured hashrate after a short test
gpu-01	GPU	KAWPOW	220	4	0.182	Use actual wall power instead of TDP when available
asic-01	ASIC	kHeavyHash	3080	0	0.182	Only run if operator already owns ASIC and power plan is verified

Coin Lanes

Coin	Algorithm	Hardware	Pool or Mode	Default posture
Monero	RandomX	CPU	Pool or P2Pool	Good learning lane; profit depends on power.
Ravencoin	KAWPOW	GPU	Public pool	GPU test lane; heat and tuning matter.
Kaspa	kHeavyHash	ASIC	ASIC pool only	ASIC-only; skip without owned compatible machine.
Ethereum Classic	Etchash	GPU	Public pool	Optional GPU comparison lane.
Bitcoin	SHA-256	ASIC	Not recommended for home CPU reference	Reference only for home labs.

Profitability Logic

Gross revenue equals current expected coin per day multiplied by current coin price. Power cost equals adjusted watts divided by 1000, multiplied by hours per day and local kWh rate. Net estimate subtracts power cost and pool fee. All live hashrate, reward, and price fields must be refreshed before each decision.

Metric	Formula
Gross revenue	$\text{gross_coin_per_day} * \text{coin_price_usd}$
Power cost	$((\text{device_watts} * (1 + \text{overhead_pct})) / 1000) * \text{hours_per_day} * \text{kwh_rate_usd}$
Pool fee	$\text{gross_revenue_usd} * \text{pool_fee_pct}$
Net estimate	$\text{gross_revenue_usd} - \text{power_cost_usd} - \text{pool_fee_usd}$

Operating Workflow

- Fill hardware intake from local inventory commands and actual utility rate.
- Create receive-only wallet addresses and update the wallet matrix.
- Refresh calculator outputs, price, difficulty, pool fee, and measured power.
- Run a short observed test with a known stop condition.
- Record accepted/rejected shares, temperature, wall power, payout, and transaction IDs.
- Append sell-off rows only after actual conversion receipts exist.

Sell-Off Controls

The sell-off path is deliberately small and auditable: confirm ownership, verify exchange support for the exact network, send a test transfer when practical, record transaction IDs, wait for confirmations, and update project cash only after settlement.

Step	Evidence to keep
Pre-transfer	Wallet balance, source address, exchange deposit network
Broadcast	Transaction ID, timestamp, amount, fee
Exchange	Deposit confirmation, order ID, gross proceeds, fees
Ledger	Append-only receipt row and correction row if needed

Source Register

Sources were refreshed on 2026-05-29 for algorithm and electricity context. Live profitability must still be recalculated at run time because price, difficulty, pool luck, fees, and wall power move continuously.

Source	Used for	URL
RandomX Moneropedia	Monero CPU-focused RandomX description	https://web.getmonero.org/resources/moneropedia/randomx.html
Monero Tail Emission	Monero tail emission and block reward boundary	https://www.getmonero.org/resources/moneropedia/tail-emission.html
Ravencoin About	Ravencoin KAWPOW and GPU hardware framing	https://ravencoin.org/about/
Ravencoin Pools	Ravencoin pool workflow and wallet-first setup	https://ravencoin.org/pools/
Kaspa Home	Kaspa kHeavyHash consensus algorithm	https://kaspa.org/
Kaspa Emission Schedule	Kaspa emission schedule	https://kaspa.org/wp-content/uploads/2022/09/KASPA-EMISSION-SCHEDULE.pdf
EIA Short-Term Energy Outlook Electricity	US residential electricity price context	https://www.eia.gov/outlooks/steo/report/elec_coal_renew.php
EIA Electricity Data	State and sector electricity data refresh path	https://www.eia.gov/electricity/data.php
XMRig Pool Configuration	XMRig pool config fields	https://xmrig.com/docs/miner/config/pool
XMRig GitHub	XMRig releases and source review	https://github.com/xmrig/xmrig
kawpowminer GitHub	Ravencoin open-source miner option	https://github.com/RavenCommunity/kawpowminer

Final Operator Checklist

- Every wallet row has receive-address status, not private-key material.
- Every device has owner, location, breaker/circuit, cooling status, and measured or conservative wattage.
- Every mining run has a stop condition for heat, power, or negative expected value.
- Every payout has a pool receipt or transaction ID.
- Every conversion has exchange receipt evidence before any USD ledger update.
- Every profitability claim is timestamped and source-linked.